Tae Soo Kim

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taesookim.com

RESEARCH INTERESTS

My research connects **human-computer interaction** and **natural language processing**, focusing on AI that **helps people discover goals worth pursuing**. I design interactive systems for exploring and evaluating possibilities, and model how human goals take shape through interaction to train and evaluate AI that supports this process.

RESEARCH EXPERIENCE

02/2026–Present **SkillBench / KAIST**† — Daejeon, Republic of Korea
Research Scientist, SkillBench · Postdoctoral Researcher, KAIST
†*Alternative military service (Technical Research Personnel program).*

07/2022–10/2022 **Allen Institute for AI** — Seattle, WA, USA
Research Scientist Intern, Semantic Scholar
Mentors: Joseph Chee Chang, Amy X. Zhang, Jonathan Bragg

EDUCATION

KAIST — Daejeon, Republic of Korea

2022–2026 Ph.D. in Computer Science (*CoE Ph.D. Dissertation Award*)
Thesis: *"Disentangling Language for Interactive AI Alignment"*
Advisor: Juho Kim

2020–2022 M.S. in Computer Science (*SoC Outstanding Thesis Award*)

2015–2020 B.S. in Computer Science (*Magna Cum Laude*)

HONORS AND AWARDS

2026 CoE Ph.D. Dissertation Award — College of Engineering, KAIST (one of 16 recipients)

2026 Outstanding Ph.D. Thesis Award — School of Computing, KAIST (one of 12 recipients)

2026, 2024, 2022 Honorable Mention Award × 3 — ACM CHI

2023 Outstanding TA Award — School of Computing, KAIST

2022 AI2 Outstanding Intern of the Year Award — Allen Institute for AI (one of 4 recipients)

2022 Outstanding M.S. Thesis Award — School of Computing, KAIST

PUBLICATIONS (* indicates equal contribution)

Under Review

[U3] **SWE-WhatIf: Simulating Real Users of Coding Agents from Recorded Sessions**
Tae Soo Kim, Tongshuang Wu, Juho Kim. *Under review, 2026*

[U2] **DocuTeam: Mixed-Initiative Multi-Agent Discussions around Evolving Documents**
Heechan Lee, Juhyeon Choi, **Tae Soo Kim**, Juho Kim, Joseph Seering. *Under review, 2026*

[U1] **"When to Hand Off, When to Work Together": Expanding Human-Agent Co-Creative Collaboration through Concurrent Interaction**
Kihoon Son, Hyewon Lee, DaEun Choi, Yoonsu Kim, **Tae Soo Kim**, Yoonjoo Lee, John Joon Young Chung, HyunJoon Jung, Juho Kim. *Under review, 2026*

Conference and Journal Papers

[C18] **KnowSim: Evaluating Information Calibration in LLM Assistants with User Simulators that Learn**
Yoonjoo Lee, Hyoungwook Jin, **Tae Soo Kim**, Shaoyang Zhang, Philippe Laban, Q. Vera Liao. *EMNLP 2026 (Main Conference, to appear)*

[C17] **DiscoverLLM: From Executing Intents to Discovering Them**
Tae Soo Kim, Yoonjoo Lee, Jaesang Yu, John Joon Young Chung, Juho Kim. *ICML 2026*

[C16] **Evalet: Evaluating Large Language Models by Fragmenting Outputs into Functions**
Tae Soo Kim*, Heechan Lee*, Yoonjoo Lee, Joseph Seering, Juho Kim. *CHI 2026* — Honorable Mention Award

[C15] **ClearFairy: Capturing Creative Workflows through Decision Structuring, In-Situ Questioning, and Rationale Inference**
Kihoon Son, DaEun Choi, **Tae Soo Kim**, Young-Ho Kim, Sangdoo Yun, Juho Kim. *CHI 2026*

[C14] **CUPID: Evaluating Personalized and Contextualized Alignment of LLMs from Interactions**
Tae Soo Kim, Yoonjoo Lee, Yoonah Park, Jiho Kim, Young-Ho Kim, Juho Kim. *COLM 2025*

[C13] **One vs. Many: Comprehending Accurate Information from Multiple Erroneous and Inconsistent AI Generations**
Yoonjoo Lee, Kihoon Son, **Tae Soo Kim**, Jisu Kim, John Joon Young Chung, Eytan Adar, Juho Kim. *FAccT 2024*

[C12] **EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria**
Tae Soo Kim, Yoonjoo Lee, Jamin Shin, Young-Ho Kim, Juho Kim. *CHI 2024*

[C11] **GenQuery: Supporting Expressive Visual Search with Generative Models**
Kihoon Son, DaEun Choi, **Tae Soo Kim**, Young-Ho Kim, Juho Kim. *CHI 2024*

[C10] **Demystifying Tacit Knowledge in Graphic Design: Characteristics, Instances, Approaches, and Guidelines**
Kihoon Son, DaEun Choi, **Tae Soo Kim**, Juho Kim. *CHI 2024* — Honorable Mention Award

[C9] **The Semantic Reader Project: Augmenting Scholarly Documents through AI-Powered Interactive Reading Interfaces**
Kyle Lo, Joseph Chee Chang, Andrew Head, Jonathan Bragg, Amy X. Zhang, Cassidy Trier, (+ 47 authors including **Tae Soo Kim**), Marti A. Hearst, Daniel S. Weld. *Communications of the ACM, 2024*

[C8] **Cells, Generators, and Lenses: Design Framework for Object-Oriented Interaction with Large Language Models**
Tae Soo Kim, Yoonjoo Lee, Minsuk Chang, Juho Kim. *UIST 2023*

[C7] **Papeos: Augmenting Research Papers with Talk Videos**
Tae Soo Kim, Matt Latzke, Jonathan Bragg, Amy X. Zhang, Joseph Chee Chang. *UIST 2023*

[C6] **DAPIE: Interactive Step-by-Step Explanatory Dialogues to Answer Children's Why and How Questions**
Yoonjoo Lee, **Tae Soo Kim**, Sungdong Kim, Yohan Yun, Juho Kim. *CHI 2023*

[C5] **Stylette: Styling the Web with Natural Language**
Tae Soo Kim, DaEun Choi, Yoonseo Choi, Juho Kim. *CHI 2022* — Honorable Mention Award

[C4] **Promptiverse: Scalable Generation of Scaffolding Prompts Through Human-AI Hybrid Knowledge Graph Annotation**
Yoonjoo Lee, John Joon Young Chung, **Tae Soo Kim**, Jean Y Song, Juho Kim. *CHI 2022*

[C3] **Winder: Linking Speech and Visual Objects to Support Communication in Asynchronous Collaboration**
Tae Soo Kim, Seungsu Kim, Yoonseo Choi, Juho Kim. *CHI 2021*

[C2] **Supporting Collaborative Sequencing for Small Groups through Visual Awareness**
Tae Soo Kim, Nitesh Goyal, Jeongyeon Kim, Juho Kim, Sungsoo (Ray) Hong. *PACM HCI, 5 (CSCW), 2021*

[C1] **Design for Collaborative Information-Seeking: Understanding User Challenges and Deploying Collaborative Dynamic Queries**
Sungsoo (Ray) Hong, Minhyang (Mia) Suh, **Tae Soo Kim**, Irina Smoke, Sangwha Sien, Janet Ng, Mark Zachry, Juho Kim. *PACM HCI, 3 (CSCW), 2019*

Posters, Demos, and Workshop Papers

[D10] **Revibing HCI: Agentic Reimplementation as Community Practice**
Yoonjoo Lee, **Tae Soo Kim**, Kevin Pu, Mira Dontcheva, Bjoern Hartmann, Toby Jia-Jun Li, Brad A. Myers, Jeffrey Nichols, April Yi Wang, Xuhai Xu, Eytan Adar. *UIST 2026 Adjunct, Workshop Proposal (to appear)*

[D9] **On the Regulatory Potential of User Interfaces for AI Agent Governance**
K. J. Kevin Feng*, **Tae Soo Kim***, Rock Yuren Pang*, Faria Huq, Tal August, Amy X. Zhang. *NeurIPS 2025 Workshop on Regulatable ML*

[D8] **Interacting with AI by Manipulating Intents**
Tae Soo Kim. *CHI 2025 In-Person Doctoral Consortium*

[D7] **Evaligner: Automatic Prompt and Criteria Refinement from User Feedback**
Heechan Lee, **Tae Soo Kim**, Juho Kim. *CHI 2025 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)*

[D6] **How to Reflect Diverse People's Perspectives in Large-Scale LLM-based Evaluations?**
Yoonjoo Lee, **Tae Soo Kim**, Juho Kim. *CHI 2024 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)*

[D5] **Unveiling Disparities in Web Task Handling Between Human and Web Agent**
Kihoon Son, Jinhyeon Kwon, DaEun Choi, **Tae Soo Kim**, Young-Ho Kim, Sangdoo Yun, Juho Kim. *CHI 2024 Workshop on Computational Methodologies for Understanding, Automating, and Evaluating User Interfaces*

[D4] **LMCanvas: Object-Oriented Interaction to Personalize Large Language Model-Powered Writing Environments**
Tae Soo Kim, Arghya Sarkar, Yoonjoo Lee, Minsuk Chang, Juho Kim. *CHI 2023 Workshop on Generative AI and HCI*

[D3] **Interactive Children's Story Rewriting Through Parent-Children Interaction**
Yoonjoo Lee, **Tae Soo Kim**, Minsuk Chang, Juho Kim. *ACL 2022 Workshop on Intelligent and Interactive Writing Assistant (In2Writing)*

[D2] **Improving Video Interfaces by Presenting Informational Units of Videos**
Saelyne Yang, Sangkyung Kwak, **Tae Soo Kim**, Juho Kim. *CHI 2022 Workshop on Computational Approaches for Understanding, Generating, and Adapting User Interfaces*

[D1] **Consensus Building in Collaborative Sequencing with Visual Awareness**
Tae Soo Kim, Sungsoo (Ray) Hong, Nitesh Goyal, Jeongyeon Kim, Juho Kim. *CHI 2020 Extended Abstract*

RESEARCH MENTORING

2026–Present **Jiho Kim**, SNU UG, then Columbia University MS — Collaborative adaptive UI
2026 **Chanyoung Park**, SNU UG — LLM tutor evaluation
2025–Present **Jaesang Yu**, KAIST MS — Goal discovery [C17] and hierarchical text-to-image scaffolding
2025 **Daho Jung**, KAIST UG — LLM tutor evaluation
2024–Present **Heechan Lee**, KAIST UG, then MS — LLM evaluation and refinement [C16, D7]
2024 **Jiho Kim**, Calvin University UG — Personalized LLM alignment [C14]
2024 **Yoonah Park**, SNU UG — Personalized LLM alignment [C14]
2024 **Yuewen Yang**, Cornell Tech MS — Personalized LLM alignment
2023 **Arghya Sarkar**, NYU UG — AI writing interfaces [D4]
2021 **Junho Myung**, KAIST UG — Group-discussion moderation
2021 **DaEun Choi**, KAIST UG — Natural-language web styling [C5]
2020 **HyeonJeong Ha**, KAIST UG — Group-discussion moderation
2020 **Seungsu Kim**, KAIST UG — Multimodal collaboration [C3]

TEACHING

Teaching Assistant, KAIST

Fall 2024	**Data Structure (CS206)** Instructor: Eun Young Moon
Fall 2023	**Human-AI Interaction (CS496G)** Instructor: Juho Kim
Spring 2023	**Data Structure (CS206)** Instructor: Eun Young Moon
Spring 2022	**Design and Analysis of Algorithms (CS500)** Instructor: Sunghee Choi
Spring 2021	**Introduction to Human-Computer Interaction (CS374)** Instructor: Juho Kim
Fall 2020	**Introduction to Social Computing (CS473)** Instructor: Juho Kim

ACADEMIC SERVICE

Workshop Organizer

UIST 2026	Revibing HCI: Agentic Reimplementation as Community Practice [D10]

Program Committee

2026	ACM UIST, Associate Chair (AC)
2023–2024	ACM CHI LBW, Associate Chair (AC)

Reviewer

2026	ACL Rolling Review, May
2022–2026	ACM CHI (Outstanding Review × 8)
2023, 2025, 2026	ACM DIS
2026	ACM C&C (Outstanding Review × 1)
2022–2025	ACM UIST (Outstanding Review × 1)
2020, 2023–2025	ACM CSCW (Outstanding Review × 2)
2021, 2025, 2026	ACM CHI LBW

Student Volunteer

2021, 2023	ACM UIST
2022	ACM CHI

INVITED TALKS

	Disentangling Text for Interactive AI Alignment
04/2026	NYU Abu Dhabi · Guest lecture (CDAD-UH 1044Q)
10/2025	MILA · Human-Centered AI Reading Group
	Data Use in LLMs and Human-Centered Evaluation
12/2024	KAIST · Guest lecture (CS475)
	Interactively Evaluating LLMs on Specific, Subjective Criteria
04/2024, 12/2023	NYU Abu Dhabi · Guest lecture (CDAD-UH 1044Q)
	Talk on the "Cells, Generators, and Lenses" and "EvalLM" papers
11/2023	NAVER Tech Talk
	Designing for Object-Oriented Interaction in Large Language Model-Powered Writing Interfaces
09/2023	PSU · Guest lecture (IST 597-001)
	Mapping Multiple Modalities to Empower Novice Designers
06/2022	Manila ACM SIGCHI Chapter